```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Exercise Pandas

For these exercices we are using a dataset provided by Airbnb for a Kaggle competition. It describes its offer for New York City in 2019, including types of appartments, price, location etc.

1. Create a dataframe

Create a dataframe of a few lines with objects and their poperties (e.g fruits, their weight and colour). Calculate the mean of your Dataframe.

```
import pandas as pd

# Create a small dataframe
data = {
    'fruit': ['Apple', 'Banana', 'Orange', 'Strawberry'],
    'weight_g': [150, 120, 200, 25],
    'color_score': [8, 9, 7, 10]
}
df_fruits = pd.DataFrame(data)

# Calculate the mean of numeric columns
fruit_mean = df_fruits.select_dtypes(include=[np.number]).mean()

display(df_fruits)
print("\nMean values:")
print(fruit_mean)
```

	fruit	weight_g	color_score
0	Apple	150	8
1	Banana	120	9
2	Orange	200	7
3	Strawberry	25	10

```
Mean values:
weight_g       123.75
color_score      8.50
dtype: float64
```

2. Import

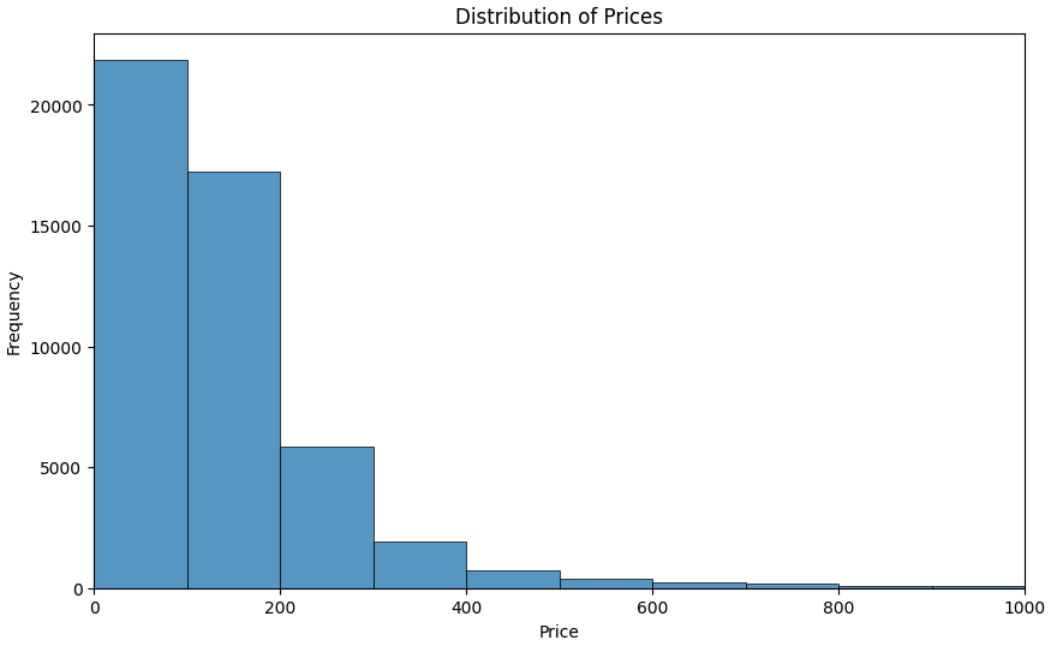
- Import the table called `AB_NYC_2019.csv` as a dataframe. It is located in the Datasets folder. Have a look at the beginning of the table (head).
- Create a histogram of prices

```
# Import the Airbnb dataset
df = pd.read_csv('/content/AB_NYC_2019.csv')

# Display the first few rows
display(df.head())

# Create a histogram of prices
plt.figure(figsize=(10, 6))
sns.histplot(df['price'], bins=100, kde=False)
plt.title('Distribution of Prices')
plt.xlabel('Price')
plt.ylabel('Frequency')
# Limiting x-axis for better visibility as there are outliers
plt.xlim(0, 1000)
plt.show()
```

	id	name	host_id	host_name	neighbourhood_group	neighbourhood	latitude	longitude	room_type	price	minimum_nights	number_of_reviews	last_
0	2539	Clean & quiet apt home by the park	2787	John	Brooklyn	Kensington	40.64749	-73.97237	Private room	149	1	9	2018
1	2595	Skylit Midtown Castle	2845	Jennifer	Manhattan	Midtown	40.75362	-73.98377	Entire home/apt	225	1	45	2019
2	3647	THE VILLAGE OF HARLEM....NEW YORK !	4632	Elisabeth	Manhattan	Harlem	40.80902	-73.94190	Private room	150	3	0	
3	3831	Cozy Entire Floor of Brownstone	4869	LisaRoxanne	Brooklyn	Clinton Hill	40.68514	-73.95976	Entire home/apt	89	1	270	2019
4	5022	Entire Apt: Spacious Studio/Loft by central park	7192	Laura	Manhattan	East Harlem	40.79851	-73.94399	Entire home/apt	80	10	9	2018



3. Operations

Create a new column in the dataframe by multiplying the "price" and "availability_365" columns to get an estimate of the maximum yearly income.

```
# 3. Operations: Create a yearly_income column
df['yearly_income'] = df['price'] * df['availability_365']
print(f"Yearly income column created. Max value: {df['yearly_income'].max()}")
```

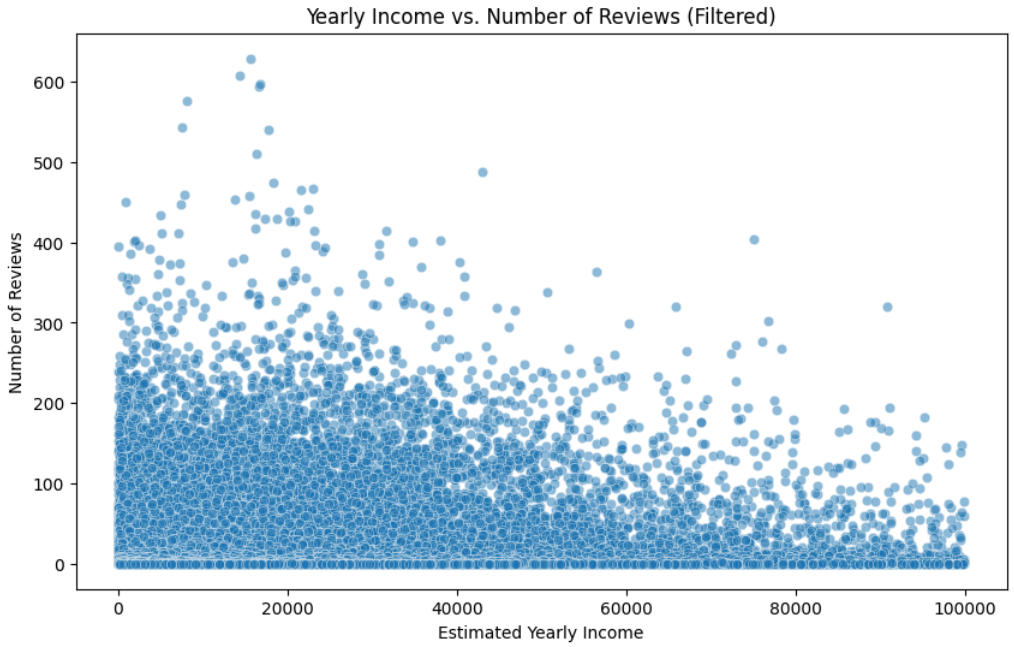
```
Yearly income column created. Max value: 3649635
```

3b. Subselection and plotting

Create a new Dataframe by first subselecting yearly incomes between 1 and 100'000. Then make a scatter plot of yearly income versus number of reviews

```
# 3b. Subselection and plotting
# Filter for yearly incomes between 1 and 100,000
df_filtered = df[(df['yearly_income'] > 0) & (df['yearly_income'] <= 100000)].copy()

# Create a scatter plot
plt.figure(figsize=(10, 6))
sns.scatterplot(data=df_filtered, x='yearly_income', y='number_of_reviews', alpha=0.5)
plt.title('Yearly Income vs. Number of Reviews (Filtered)')
plt.xlabel('Estimated Yearly Income')
plt.ylabel('Number of Reviews')
plt.show()
```



4. Combine

We provide below and additional table that contains the number of inhabitants of each of New York's boroughs ("neighbourhood_group" in the table). Use `merge` to add this population information to each element in the original dataframe.

```
# Exercise 4: Combine with population data
pop_data = {
    'neighbourhood_group': ['Manhattan', 'Brooklyn', 'Queens', 'Bronx', 'Staten Island'],
    'population': [1628706, 2559903, 2253858, 1418207, 476143]
}
df_pop = pd.DataFrame(pop_data)

# Merge population data into the original dataframe
df = df.merge(df_pop, on='neighbourhood_group', how='left')
display(df.head())
```

	id	name	host_id	host_name	neighbourhood_group	neighbourhood	latitude	longitude	room_type	price	minimum_nights	number_of_reviews	last_
0	2539	Clean & quiet apt home by the park	2787	John	Brooklyn	Kensington	40.64749	-73.97237	Private room	149	1	9	201
1	2595	Skylit Midtown Castle	2845	Jennifer	Manhattan	Midtown	40.75362	-73.98377	Entire home/apt	225	1	45	201
2	3647	THE VILLAGE OF HARLEM....NEW YORK !	4632	Elisabeth	Manhattan	Harlem	40.80902	-73.94190	Private room	150	3	0	
3	3831	Cozy Entire Floor of Brownstone	4869	LisaRoxanne	Brooklyn	Clinton Hill	40.68514	-73.95976	Entire home/apt	89	1	270	201
4	5022	Entire Apt: Spacious Studio/Loft by central park	7192	Laura	Manhattan	East Harlem	40.79851	-73.94399	Entire home/apt	80	10	9	201

5. Groups

- Using `groupby` calculate the average price for each type of room (room_type) in each neighbourhood_group. What is the average price for an entire home in Brooklyn ?
- Unstack the multi-level Dataframe into a regular Dataframe with `unstack()` and create a bar plot with the resulting table

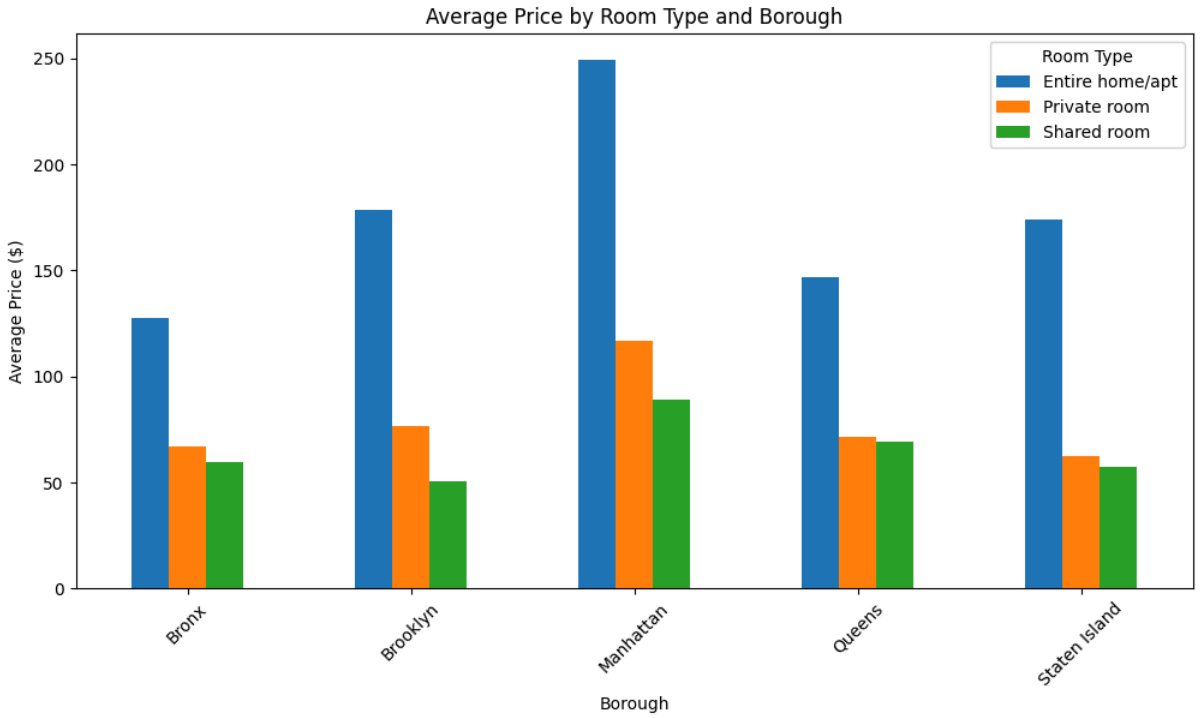
```
# Exercise 5: Groups and Unstacking
# Calculate average price for each room_type in each neighbourhood_group
avg_prices = df.groupby(['neighbourhood_group', 'room_type'])['price'].mean()

# Brooklyn average price for Entire home/apt
brooklyn_entire_home_avg = avg_prices.loc['Brooklyn', 'Entire home/apt']
print(f"Average price for an entire home in Brooklyn: ${brooklyn_entire_home_avg:.2f}")

# Unstack and plot
avg_prices_unstacked = avg_prices.unstack()
avg_prices_unstacked.plot(kind='bar', figsize=(12, 6))
```

```
plt.title('Average Price by Room Type and Borough')
plt.ylabel('Average Price ($)')
plt.xlabel('Borough')
plt.xticks(rotation=45)
plt.legend(title='Room Type')
plt.show()
```

```
Average price for an entire home in Brooklyn: $178.33
```



6. Advanced plotting

Using Seaborn, create a scatter plot where x and y positions are longitude and lattitude, the color reflects price and the shape of the marker the borough (neighbourhood_group). Can you recognize parts of new york ? Does the map make sense ?

```
# Exercise 6: Advanced plotting
plt.figure(figsize=(12, 8))
sns.scatterplot(data=df, x='longitude', y='latitude', hue='price', style='neighbourhood_group',
                palette='viridis', alpha=0.6, size='price', sizes=(10, 200))
plt.title('Airbnb Listings in NYC by Coordinates, Price, and Borough')
plt.xlim(-74.25, -73.7)
plt.ylim(40.48, 40.92)
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left')
plt.show()
```


Airbnb Listings in NYC by Coordinates, Price, and Borough
40.90
price
0
2000